R Laboratory Session 0

March 17, 2026

Exercise 1: Vectors and Data Frames

The following table (from `https://en.wikipedia.org/wiki/List_of_lochs_of_Scotland`) reports the volume, area, length, and maximum and mean depth of several Scottish lakes:

Loch	Volume [km^3]	Area [km^2]	Length [km]	Max. depth [m]	Mean depth [m]
Loch Ness	7.45	56	39	230	132
Loch Lomond	2.6	71	36	190	37
Loch Morar	2.3	27	18.8	310	87
Loch Tay	1.6	26.4	23	150	60.6
Loch Awe	1.2	39	41	94	32
Loch Maree	1.09	28.6	20	114	38
Loch Ericht	1.08	18.6	23	156	57.6
Loch Lochy	1.07	16	16	162	70
Loch Rannoch	0.97	19	15.7	134	51
Loch Shiel	0.79	19.5	28	128	40
Loch Katrine	0.77	12.4	12.9	151	43.4
Loch Arkaig	0.75	16	19.3	109	46.5
Loch Shin	0.35	22.5	27.8	49	15.5

1. Create R vectors containing the lake names and the corresponding numerical variables. Build a data frame called `scottish.lakes`.

2. Determine:

   - the lake with the largest and the smallest volume,
   - the lake with the largest and the smallest surface area.

3. Order the data frame with respect to the lake area and determine the two lakes with the largest area.

4. By summing all the lake areas, compute the total area covered by these lakes.

Exercise 2: 100 m Athletics Records

The website

`https://www.alltime-athletics.com/m_100ok.htm`

contains a list of the best recorded times in official men's 100 m races.

1. Open the results on the Web, reported as a table and create a `tibble` that contains all the results.

2. To parse the Web data, use the `rvest` and `tidyverse` packages and the following code:

```
library(rvest)
library(tidyverse)
men100m_html <- read_html("http://www.alltime-athletics.com/m_100ok.htm")
men100m_html |> html_nodes(xpath = "//pre") |> html_text() -> men100m_list
men100m_tbl <- read_fwf(men100m_list)
```

3. The resulting `tibble` has all data represented as "character". Manipulate the `tibble` to convert the columns to the proper format.

4. Once `men100m_tbl` is properly formatted realize the following plots:

   (a) evolution of the fastest time as a function of date of the races

   (b) which country had the largest number of the fastest runners each year?

5. Repeat the same analysis for the women's 100 m races using

   `https://www.alltime-athletics.com/w_100ok.htm`

   and compare the results between men and women.

Tasks

Prepare

1. a PDF report containing
    - the commented code (copied/pasted)
    - outputs (vectors, plots, etc.)
    - short explanations of the code
2. the corresponding .R (source code).

Both files should be handed in via Moodle by Tuesday 24.

This first exercise sheet **will not be graded**.